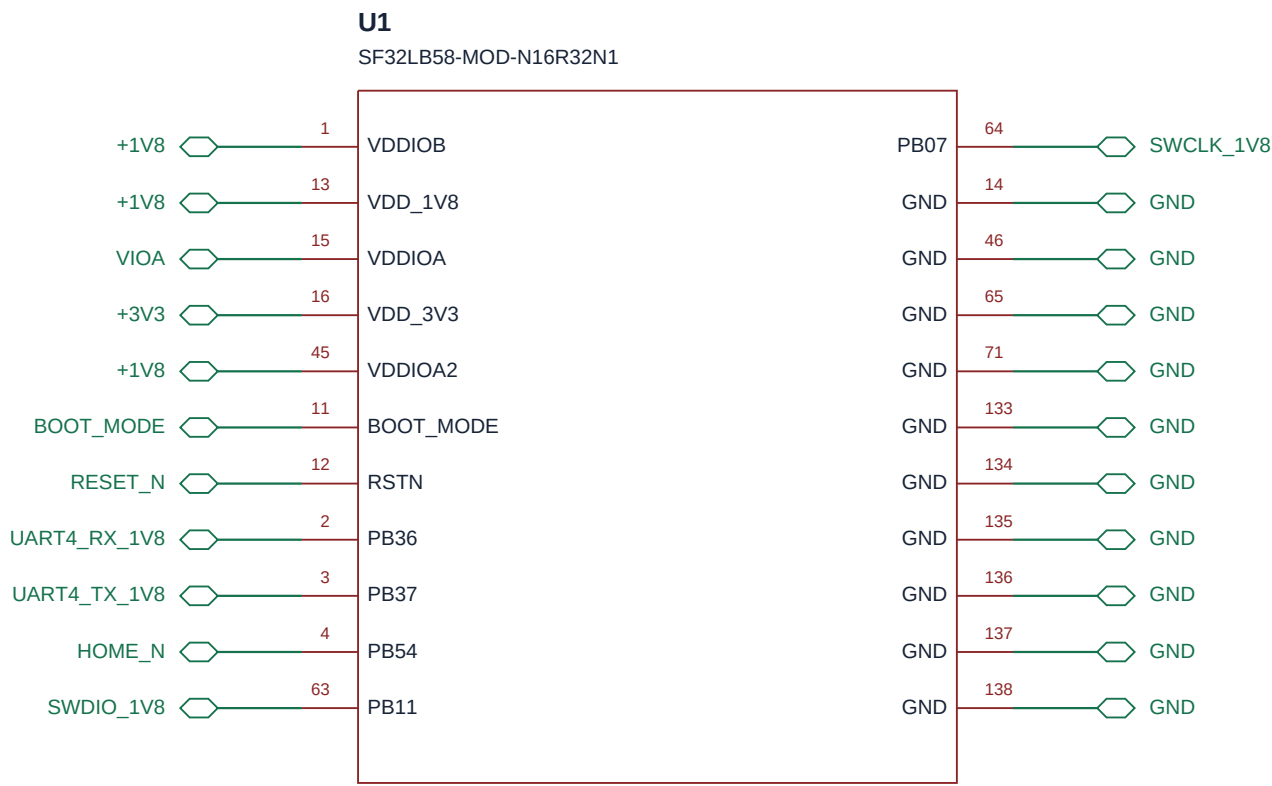


HealthWatch-SF58-Offline-v0.1 | 01 CORE AND DEBUG

SF32LB58 module carrier | PA bank selectable; PB / SWD / UART = 1.8 V

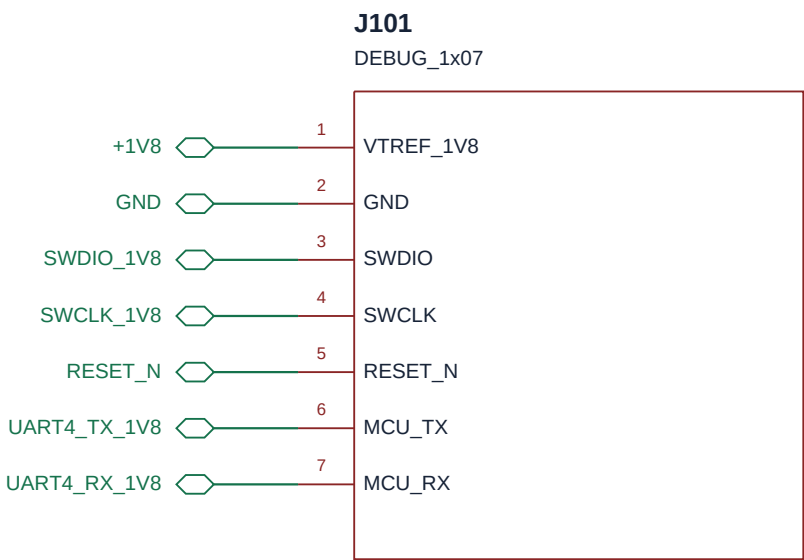


Module contains 48 MHz + 32.768 kHz crystals, PSRAM and NOR flash.

HCPU 240 MHz target; start low during bring-up. RTOS / LVGL platform.

N16R32N1: 32 MB PSRAM total, 16 MB NOR + 1 MB boot NOR.

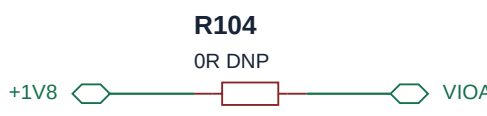
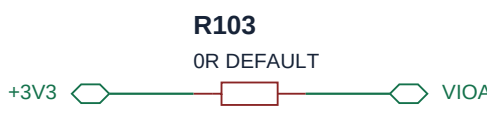
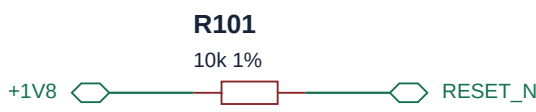
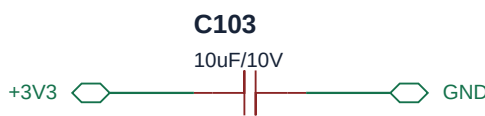
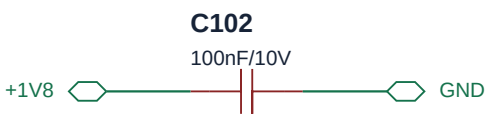
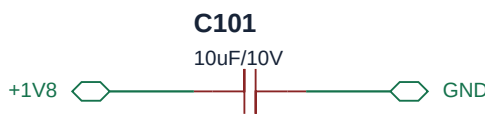
This carrier is for electrical validation; final 46 mm watch PCB is a later revision.



VTREF is a voltage reference output, not a power input.

Use a debugger / UART adapter with true 1.8 V logic. Cross TX and RX.

All unused module pads are shown on sheet 08.



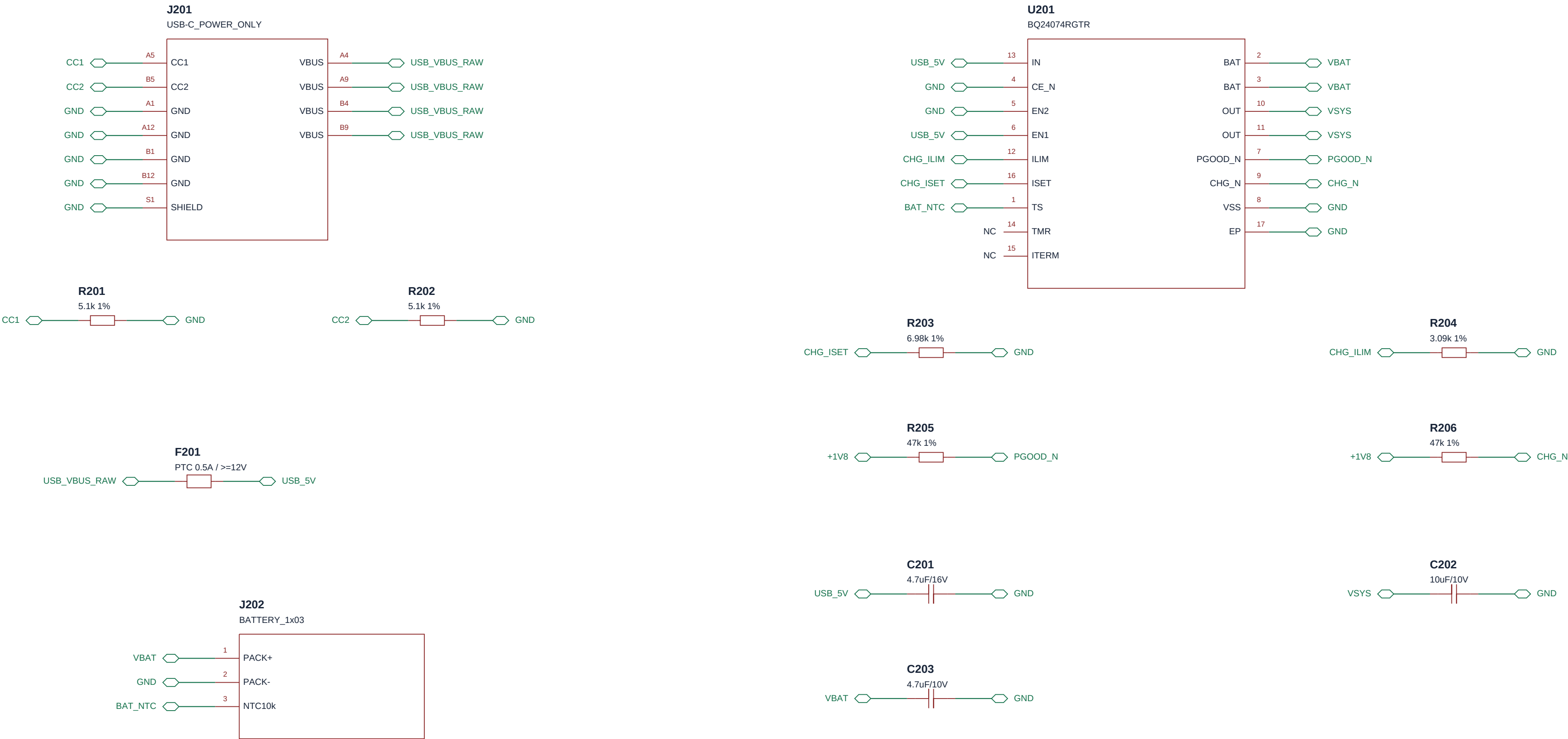
Fit R103 only: VIOA = 3.3 V for external display adapter.

For a 1.8 V panel interface: remove R103, then fit R104 and recheck pull-ups.

BOOT_MODE belongs to VIOA; RESET_N pull-up must stay on +1V8.

HealthWatch-SF58-Offline-v0.1 | 02 BATTERY AND CHARGER

BQ24074 power path | single-cell 4.2 V charge termination | nominal 128 mA initial charge current

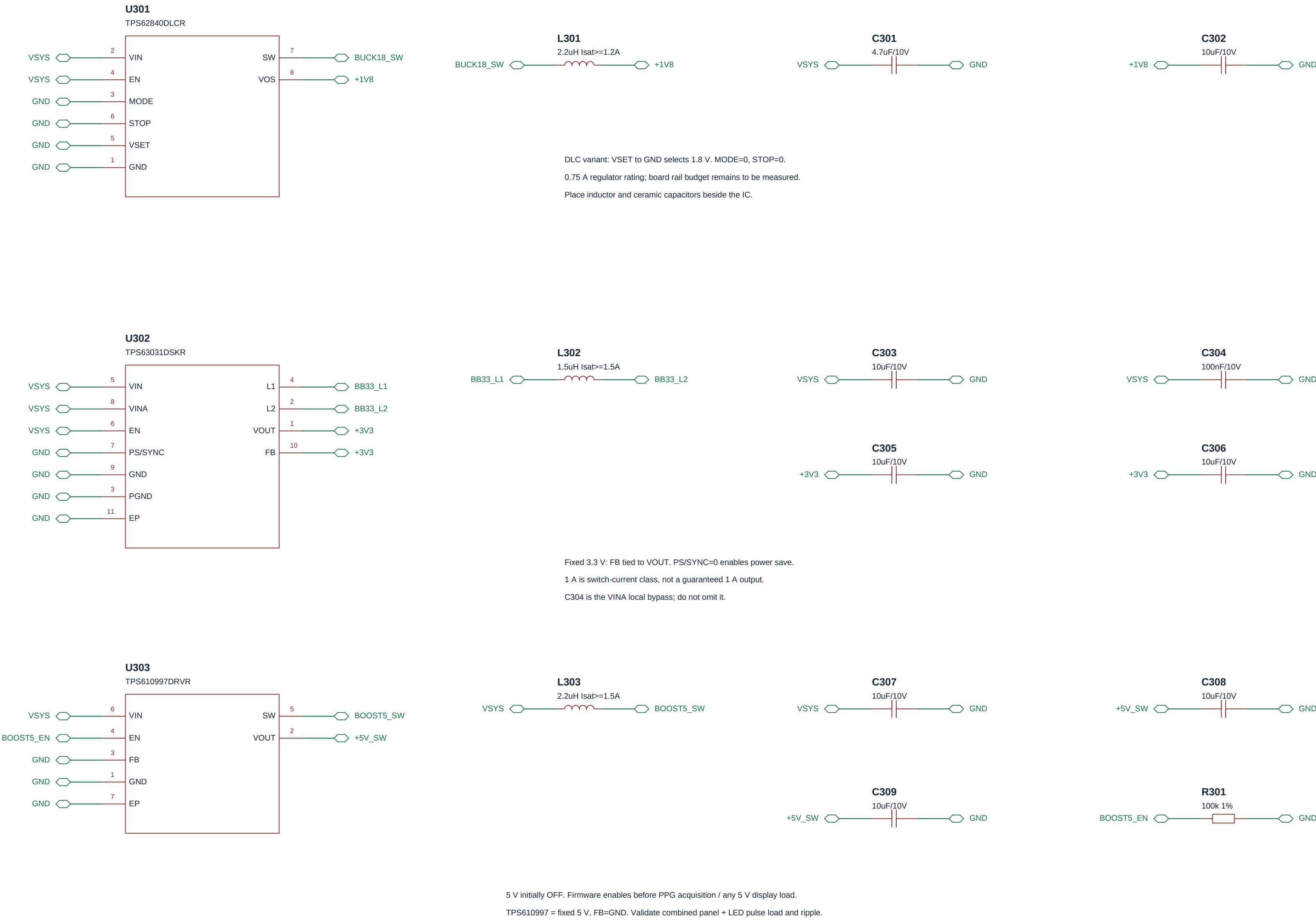


Battery: protected 1S Li-ion/LiPo, 4.2 V charge rating; nominal 3.7 V.
400 mAh is a planning assumption. Verify max charge current and cell dimensions.
Use a real 10k NTC in thermal contact with the cell; TS is not tied to GND.
Connector polarity and NTC B value must be checked against the selected pack.
USB-C is charge-only. USB data / SBU contacts are not routed in this revision.

$I_CHG\ typ = 890 / 6980 = 127.5\ mA$ (charger / resistor tolerance applies).
EN2=0, EN1=1: fixed 500 mA input mode; use a suitable USB 5 V supply.
ILIM resistor fitted as required even when fixed input mode is selected.
TMR and ITERM open: internal default timer / termination behavior.
VSYS ~ battery voltage on battery; up to 4.4 V with external input.
Protection IC is required in the purchased battery pack; not duplicated here.

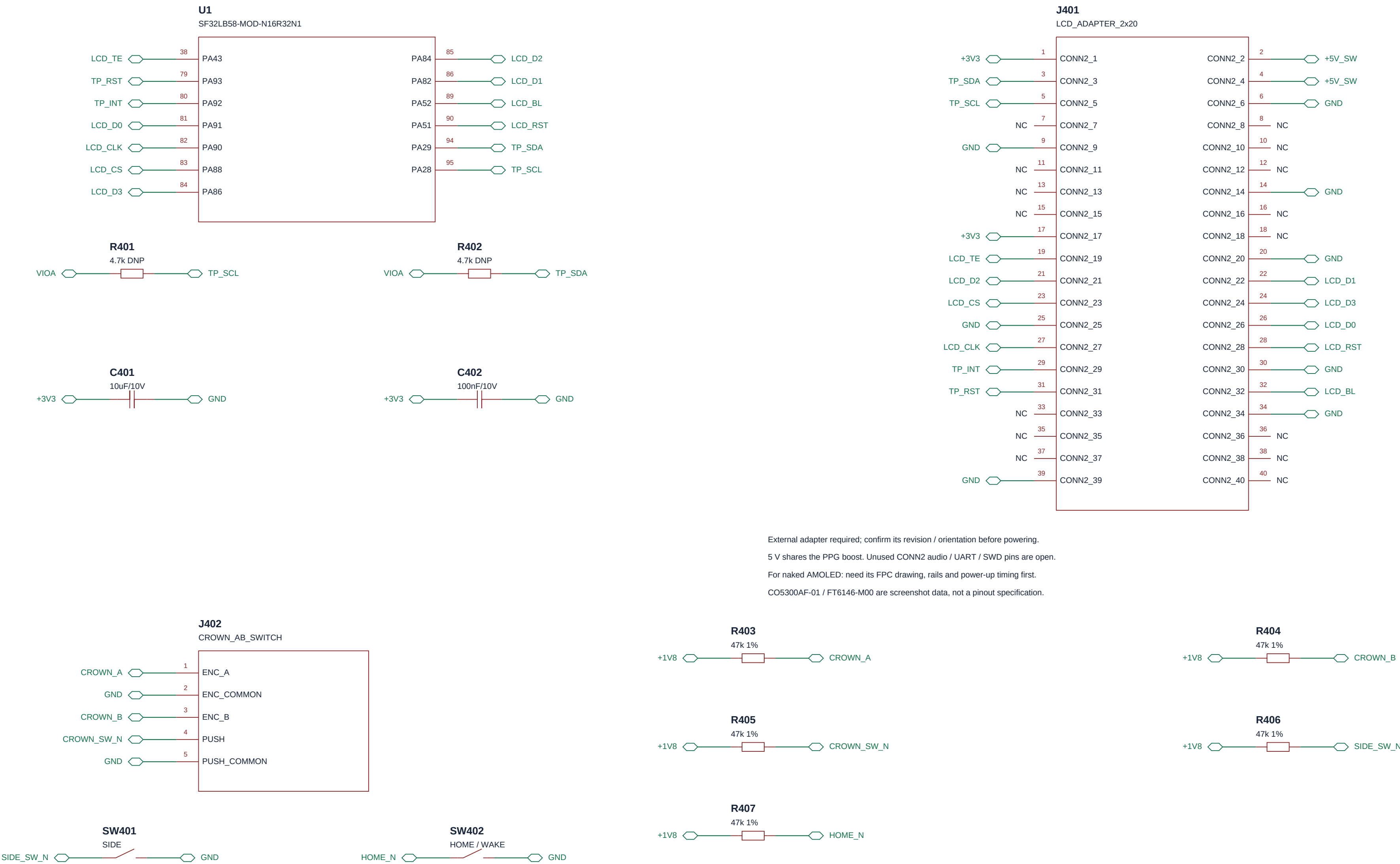
HealthWatch-SF58-Offline-v0.1 | 03 POWER RAILS

VSYS -> 1.8 V buck, 3.3 V buck-boost, switchable 5 V boost | validate startup and peak current



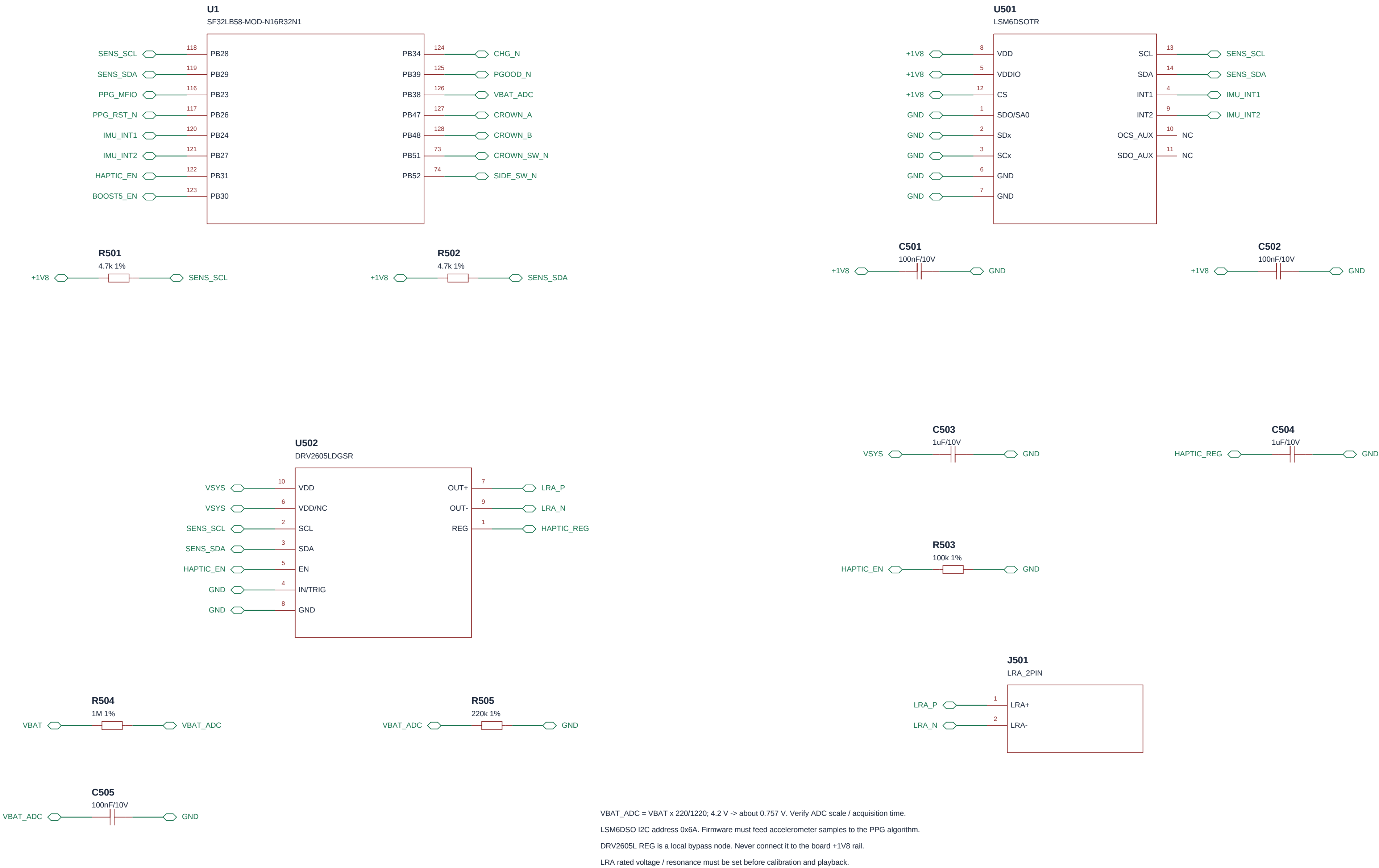
HealthWatch-SF58-Offline-v0.1 | 04 DISPLAY AND CONTROLS

390x450 QSPI AMOLED + I2C touch | connector matches required SiFiI CONN2 signals; panel FPC is not guessed



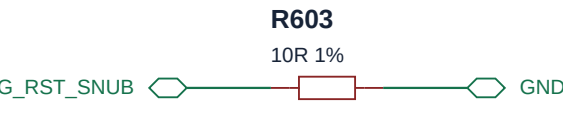
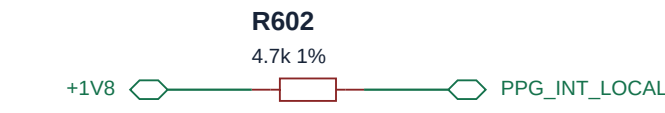
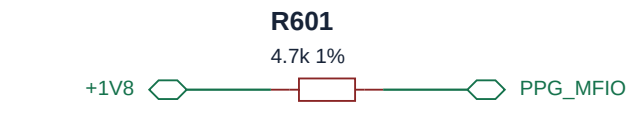
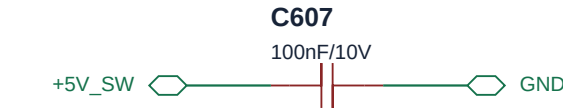
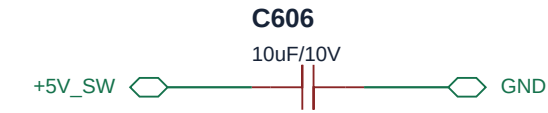
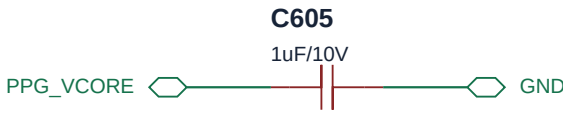
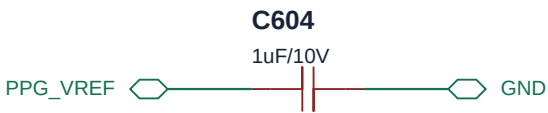
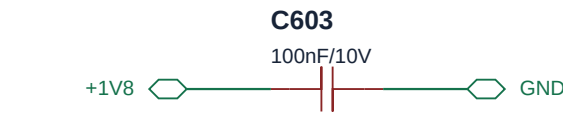
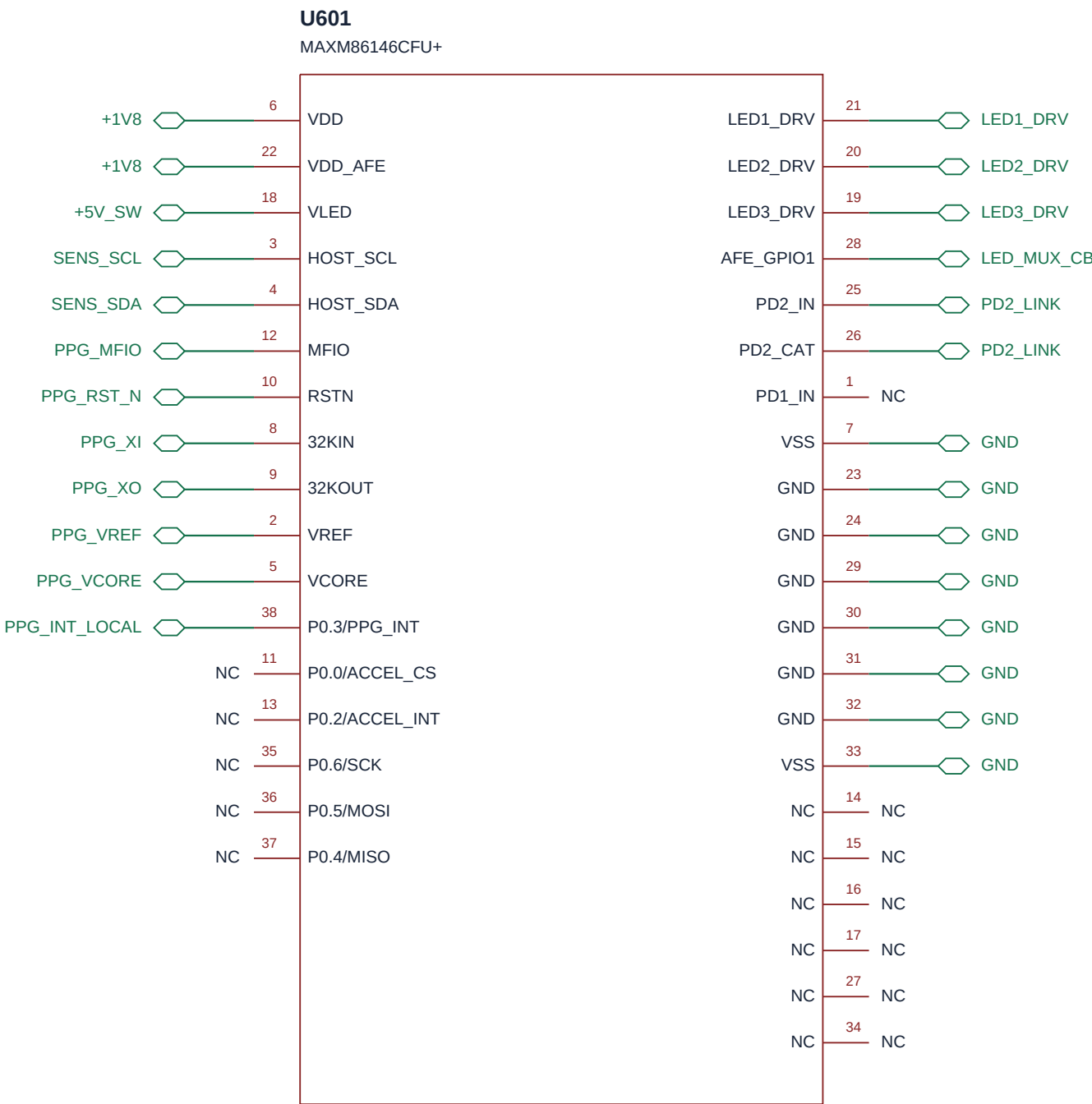
HealthWatch-SF58-Offline-v0.1 | 05 IMU AND HAPTICS

LSM6DSO -> SF58 host -> MAXM86146 algorithm | 1.8 V sensor I2C bus | LRA with closed-loop driver



HealthWatch-SF58-Offline-v0.1 | 06 PPG MODULE

MAXM86146CFU+ heart-rate / SpO2 front end | wrist optics and firmware algorithm still require validation



PD1 is integrated: pin 1 remains open. PD2 pin 25 must be joined to pin 26.

VCORE and VREF are capacitor-only nodes; do not drive with +1V8.

The unused accelerometer SPI pins stay unloaded in host-feed architecture.

PPG_INT_LOCAL is the local AFE interrupt; host communication uses MFIO.

Crystal load network follows the ADI reference; no guessed external load capacitors.

Host I2C / MFIO / reset all 1.8 V. I2C pull-ups are on sheet 05.

Hold PPG reset while required rails / clock stabilize, then follow ADI boot protocol.

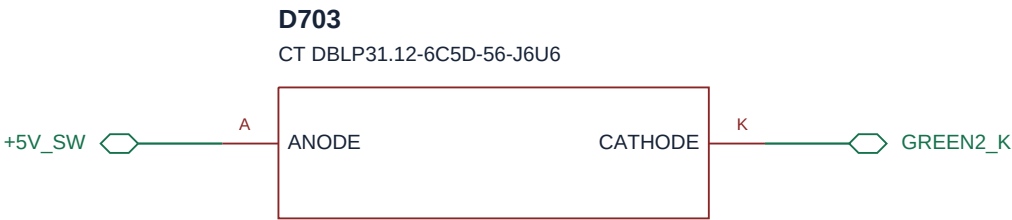
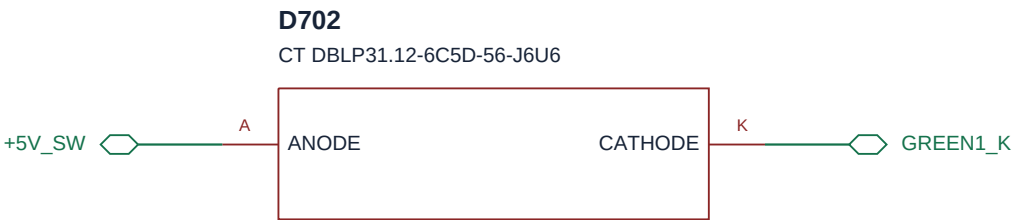
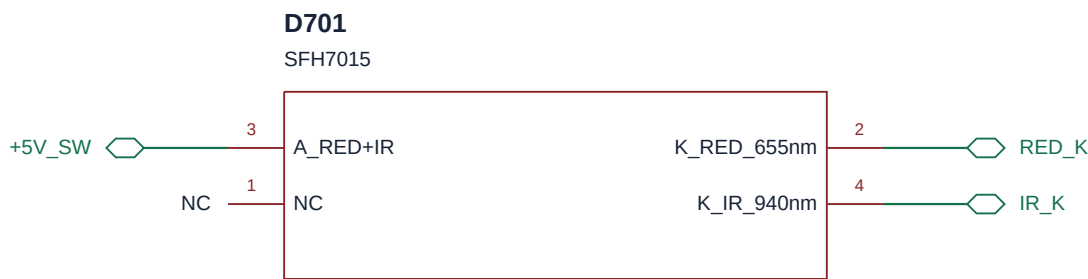
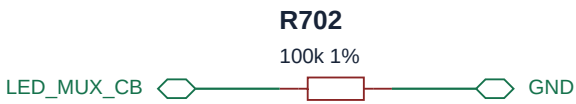
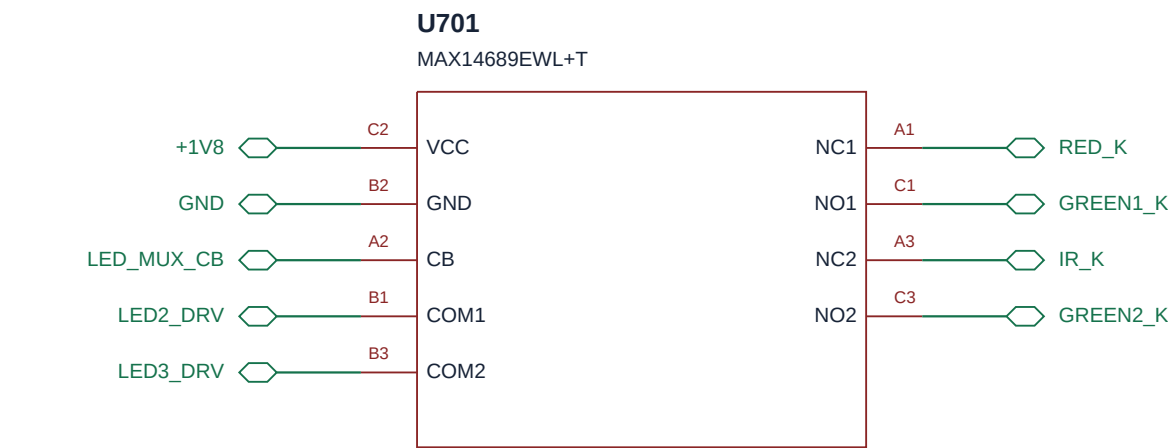
C601 at VDD; C602+C603 at VDD_AFE; C606+C607 at VLED.

Common ground electrically; keep LED pulse returns away from optical/analog nodes.

This is a real sensing circuit; accuracy is not established by the schematic.

HealthWatch-SF58-Offline-v0.1 | 07 OPTICAL EMITTERS

Two green LEDs + red / IR | MAX14689 switching network | LED polarity / optical footprint review required



Current is regulated by MAXM86146 LED drivers. Set pulse current / timing before acquisition.

CB=0: LED2 -> RED, LED3 -> IR. CB=1: LED2 -> GREEN1, LED3 -> GREEN2.

GREEN1 cathode is also connected to LED1_DRV, following the ADI typical application.

Keep the GREEN1 shared path: three independent LEDs would change the reference design.

MAX14689 VCC=1.8 V; analog terminals support beyond-rail signals within its ratings.

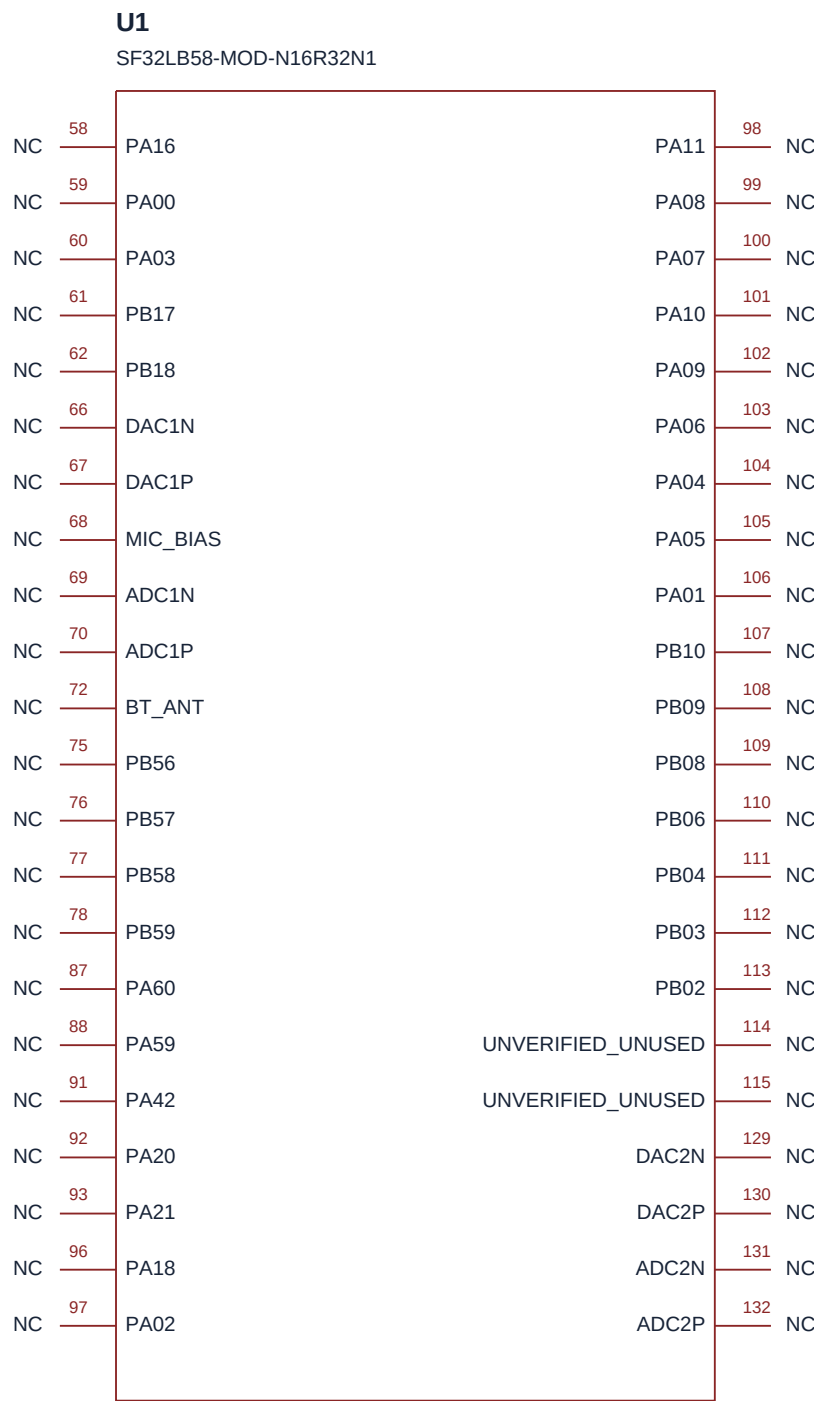
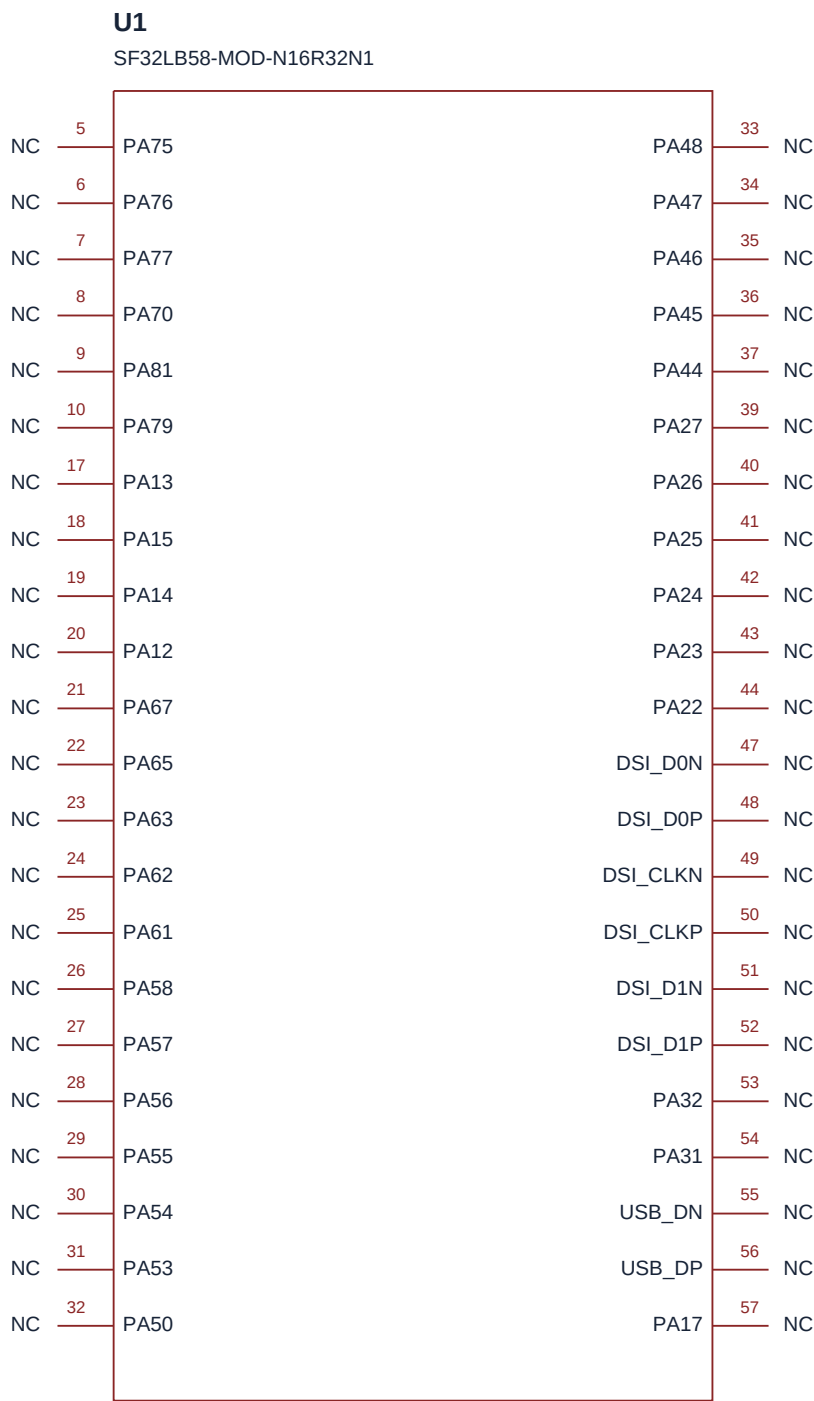
Green symbols use A/K logical pads. Physical pad mapping is intentionally not invented.

Before PCB: audit emitter land patterns, optical separation, black barriers and skin contact geometry.

Heart rate / SpO2 algorithms, calibration, motion rejection and human testing remain required.

HealthWatch-SF58-Offline-v0.1 | 08 MODULE UNUSED PADS

Complete module inventory | offline phase: MIPI, USB data, RF, audio and spare GPIO remain unconnected



U1 units A-E are the same physical SF32LB58 module. Shared designator and unique ID preserve one BOM item.

All pads marked NC on this sheet are deliberately not routed in v0.1. This is not an omission of power pins.

114 / 115 are shown as UNVERIFIED_UNUSED; confirm names using the module pad drawing before future reuse.

The imported module footprint comes from the SiFiI native EDA reference project.

Other IC and mechanical footprints are unbound for deliberate review before PCB layout.

No PCB outline, RF antenna, audio chain, network radio enablement or production release is included.

Next gate: verify imported electrical nets, complete footprint audit, confirm panel adapter and battery pack.